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DEPARTMENT OF COMPUTER ENGINEERING

Subject : Machine Learning Lab		Remark	
Name: _____ Roll No. : _____			
Class: T.Y. Comp Engg. Batch : T Division: _____		Signature	
Expt. No. : 08 Date : / /			
Title : Write a program to implement the K-means Clustering Algorithm for a sample training data set stored as a .CSV file. Compute the accuracy of the classifier, considering few test data sets and Visualize model with Python.			
Rubrics:	Date of Submission:		
	Marks Split up to	Maximum Marks	Marks Obtained
	Performance/conduction	3	
	Report Writing	3	
	Attendance	2	
	Viva/Oral	2	
	Total Marks	10	
	Aim : The main aim of this lab is to demonstrate the how to K-means Clustering with Python Secondary aim of this lab is to understand - What is K-means Clustering Algorithm? - How K-means Clustering works? - How to Choosing the right number of clusters? - How to Fit a K-means Clustering using Scikit-Learn? - How to Visualize K-means clustering using Matplotlib?		
Software Required :	Jupyter Notebook , Python 3.6, Or Google Colab		
Dataset used	Mall_customer.csv		

Theory :

Introduction to K-means clustering

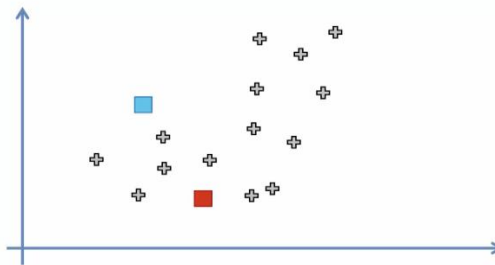
K-Means Clustering is an unsupervised machine learning algorithm. In contrast to traditional supervised machine learning algorithms, K-Means attempts to classify data without having first been trained with labeled data. Once the algorithm has been run and the groups are defined, any new data can be easily assigned to the most relevant group.

The real world applications of K-Means include:

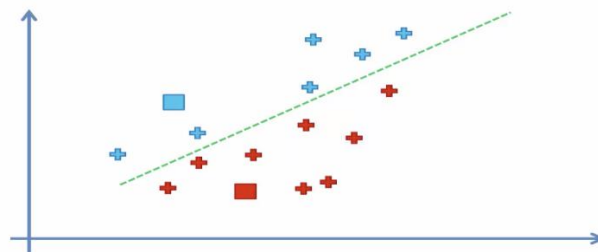
1. Customer profiling
2. Market segmentation
3. Computer vision
4. Search engines
5. Astronomy

How it works

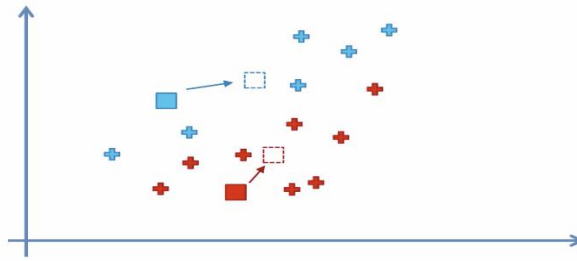
1. Select K (i.e. 2) random points as cluster centers called centroids



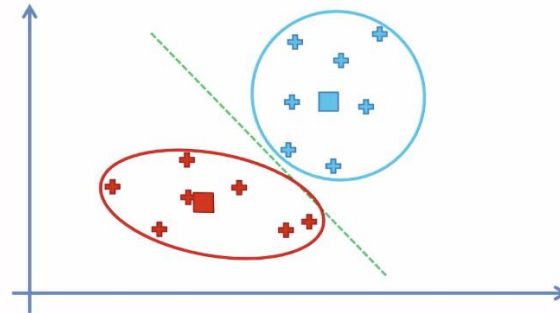
2. Assign each data point to the closest cluster by calculating its distance with respect to each centroid



3. Determine the new cluster center by computing the average of the assigned points



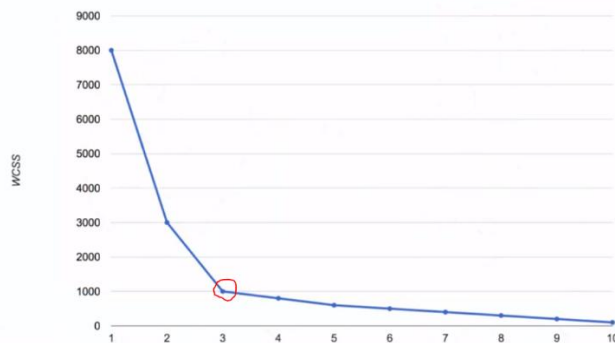
4. Repeat steps 2 and 3 until none of the cluster assignments change



Choosing the right number of clusters

Often times the data you'll be working with will have multiple dimensions making it difficult to visual. As a consequence, the optimum number of clusters is no longer obvious. Fortunately, we have a way of determining this mathematically.

We graph the relationship between the number of clusters and Within Cluster Sum of Squares (WCSS) then we select the number of clusters where the change in WCSS begins to level off (elbow method).



WCSS is defined as the sum of the squared distance between each member of the cluster and its centroid.

$$WSS = \sum_{i=1}^m (x_i - c_i)^2$$

For example, the computed WCSS for figure 1 would be greater than the WCSS calculated for figure 1.

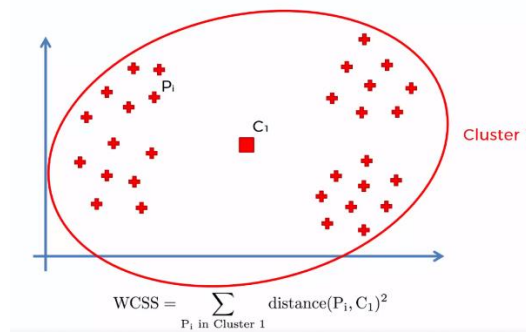
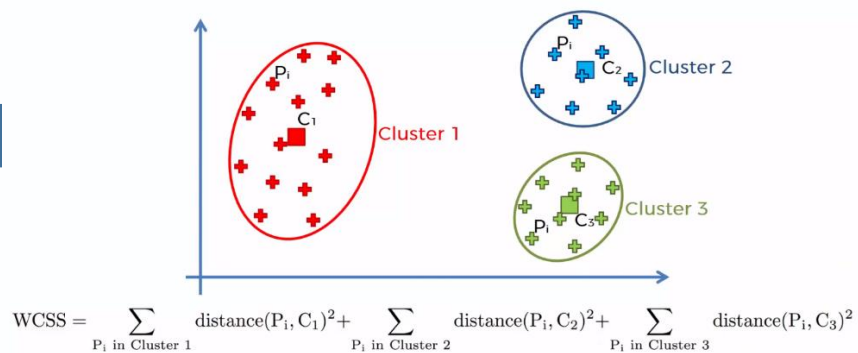


Figure 1

Figure 2



Procedure

1. Import the required libraries.
2. Import the Dataset.
3. Using the elbow method to find the optimal number of clusters

Even though we already know the optimal number of clusters, I figured we could still benefit from determining it using the elbow method. To get the values used in the graph, we train multiple models using a different number of clusters and storing the value of the **inertia_ property** (WCSS) every time.

4. Training the K-Means model on the dataset

Categorize the data using the optimum number of clusters (5) we determined in the last step. k-means++ ensures that you get don't fall into the random initialization trap

5. Generate Statistics from the initialization run with the lowest SSE are available as attributes of kmeans after calling .fit()
6. Creating Output labels for Generating Graph
7. Visualize the clusters

Practice Problems:

1. Implement K-means Clustering Algorithm
2. Visualize your build model in python

Conclusion Test Your Knowledge	In this way we successfully implement and visualize K-means Clustering Algorithm in python.	
	1. For the model we created in lab session write the following results	
	Optimal cluster Number	
	Cluster Inertia	
	Cluster Centers	
	No. of Iteration Required to converge	
	homogeneity_score	
	2. Create the model using following parameter setting and write down the result n_clusters = 5 init = random n_init = any number of your choice above 10 max_iter = 100 + your roll number tol = 0.00001 random_state = any number of your choice	
	Optimal cluster Number	
	Cluster Inertia	
Cluster Centers		
No. of Iteration Required to converge		
homogeneity_score		